Emre Ertürk

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RESEARCH INTERESTS

Evolutionary computation, artificial life, bio-inspired robotics and systems, swarm robotics, evolutionary and developmental robotics, robot locomotion, simulation and computer graphics. Also psychology, cognitive science, biology and neuroscience.

EDUCATION

B.Sc. in Computer Engineering, İzmir Institute of Technology 2022–present

CGPA 3.12 / 4.00 (fall 2025–26 term: 3.83). Senior thesis: Smart Zebra, with Onur Kansoy (equal contribution), advised by Assoc. Prof. Selma Tekir, funded by TÜBİTAK 2209-A.

Relevant coursework: Behavioral Robotics (AA), Evolutionary Computation (BA), Senior Thesis and Seminar I, II (AA, AA), Natural Language Understanding and Generation (AA), Embedded Computer Systems (BA), Analysis and Design of Algorithms (BA), Operating Systems (BA), Differential Equations (AA). **This term:** Introduction to Computer Graphics, Introduction to Image Understanding, Elementary Chinese II.

İzmir Atatürk Lisesi, İzmir High school

Member of IALHUK, the school aviation club.

PAPERS AND REPORTS

O. Kansoy, **E. Ertürk** (equal contribution). **Smart Zebra: A Prompt-to-GRPO Recipe for Logic-Grid Reasoning in Small Language Models.** *Manuscript in revision, 2026. B.Sc. thesis.*

E. Ertürk. AEASS v3: Revisiting an Adaptive Evolutionary Algorithm for Solving Sudokus. *Technical report, 2026.*

E. Ertürk. Slavemaker Swarm: an individual-based simulation of Polyergus slave raids on Formica colonies. *Project report v0.2, 2026.*

H. N. Kömürcü, D. Karabacak, **E. Ertürk**, T. İyiarı. **An Indoor Mobile Robot (P.C.B) for Corridor Navigation and Secure Delivery Using IMU-Aided Control and ArUco Landmarks.** *CENG 483 course paper, 2026.*

RESEARCH PROJECTS

EvoMaxx v10: an artificial life sandbox on the GPU 2026, ongoing

WebGL2 system in which an eight-gene regulatory network drives the red, green and blue channels of every pixel, and the world generates its own music. Mapped a 24,480-rule space with controlled experiments on about 18,000 worlds; raised the share of lifelike genomes from 7% to 95%. Reproduced twelve papers it builds on (Lenia, Flow-Lenia, Physarum, MAP-Elites and others).

Slavemaker Swarm: individual-based model of dulosis 2026

3-D agent-based simulation of Polyergus raids with five pheromone fields, an enslavement pipeline, queen usurpation, slave rebellion and colony-level coevolution.

AEASS v3: memetic algorithm for constraint satisfaction 2025; revised 2026

GA with Lamarckian min-conflicts local search and ILS escapes on logic-reduced domains, in Python, C99 and WebGL2. 65,201 of 65,201 verified runs solved over 52,458 benchmark puzzles, with a full ablation.

Ongoing reproductions 2026

Evolutionary computation and swarm robotics: BEECLUST aggregation with simulated Kilobots in ARGoS, the Vicsek ordering transition, NEAT on XOR, Gray–Scott patterns.

ROBOTICS

P.C.B indoor delivery robot, CENG 483 Behavioral Robotics Fall 2025

Team of four. Arduino Mega and Jetson Nano; IMU-aided PID heading control; reactive ultrasonic navigation; ArUco landmarks; three prototypes.

PartyBot, CENG 424 Embedded Computer Systems Spring 2026

Team of four. Gesture-controlled mecanum robot; two ESP32 boards over ESP-NOW.

SKILLS

Programming: Python, C, JavaScript/TypeScript, GLSL. **Simulation and graphics:** WebGL2, GPU compute, agent-based modelling, ARGoS. **Robotics:** Arduino, Jetson Nano, ESP32, OpenCV. **Methods:** evolutionary algorithms, local search, experimental design and statistical testing.

LANGUAGES AND TESTS

Turkish (native); English (C1, YDS 93.5); Chinese (elementary, Chinese I and II); German (learning). ALES 85.